

Submission to Heidelberg Integration Bee

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Introduction.

Hi, I'm Zye.

I am a Korean youtuber who has made a few math videos. Thank you for giving this opportunity to participate in HIB! As an expression of gratitude, I've decided to try recording my greeting video in German, with help from dictionaries and a few friends.

However, I don't speak German at all, so the task was very challenging. "*Heraus... Herausford... Herausforderung...*" I had to record every word again and again. And then I thought, **"I want to turn this experience into a math question!"**

This question might look very *foreign* to you at first. You have to go through it *again and again*, like how I recorded my German introduction. However, as I kept repeating I felt that I was getting better, that I could say *more and more syllables at once*. Maybe you can have that experience too, with this integral!

Question.

$$\int_0^{4\pi} \sum_{n=1}^{\left\lceil \frac{x}{\pi} \right\rceil} (i \sin x)^n dx$$

L^AT_EX Representation:

```
\int\limits_{0}^{4\pi}\,,  
\sum_{n=1}^{\textstyle\left\lceil\frac{x}{\pi}\right\rceil}\,,  
\left(i\sin x\right)^n\,,  
\mathrm{d}x
```

Answer.

$$-\frac{9\pi}{8}$$

L^AT_EX Representation: `-\frac{9\pi}{8}`

Solution.

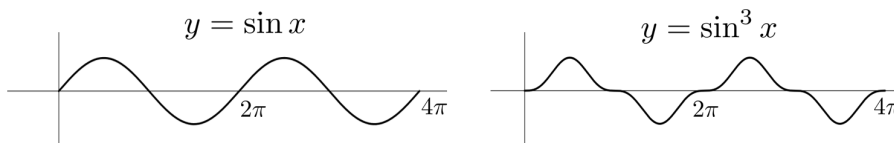
For $n \in \mathbb{N}$, the final index of the sum is **constant** if $n\pi < x < (n+1)\pi$.
Therefore, we divide the integral into 4 units.

$$\int_0^{4\pi} \sum_{n=1}^{\left\lceil \frac{x}{\pi} \right\rceil} (i \sin x)^n \, dx = \int_0^{\pi} \sum_{n=1}^1 (\dots) + \int_{\pi}^{2\pi} \sum_{n=1}^2 (\dots) + \int_{2\pi}^{3\pi} \sum_{n=1}^3 (\dots) + \int_{3\pi}^{4\pi} \sum_{n=1}^4 (\dots)$$

Look at the length of the sum slowly growing as the interval goes up!
That's a metaphor of me being able to pronounce more syllables at once, as I do more takes.
Now, note which n values appear inside which intervals, and **rearrange** the integrals.

$$\begin{aligned} & \int_0^{4\pi} (\mathbf{n=1}) + \int_{\pi}^{4\pi} (\mathbf{n=2}) + \int_{2\pi}^{4\pi} (\mathbf{n=3}) + \int_{3\pi}^{4\pi} (\mathbf{n=4}) \\ &= \int_0^{4\pi} i \sin x \, dx - \int_{\pi}^{4\pi} \sin^2 x \, dx - \int_{2\pi}^{4\pi} i \sin^3 x \, dx + \int_{3\pi}^{4\pi} \sin^4 x \, dx \\ &= - \int_{\pi}^{4\pi} \sin^2 x \, dx + \int_{3\pi}^{4\pi} \sin^4 x \, dx \end{aligned}$$

The $\sin x$ term and the $\sin^3 x$ term were removed using their symmetries.



※ Some contestants may already know that

$$\int_0^{\pi} \sin^2 x \, dx = \frac{\pi}{2} \quad \text{and} \quad \int_0^{\pi} \sin^4 x \, dx = \frac{3\pi}{8}.$$

In that case, the answer is simply

$$-3 \int_0^{\pi} \sin^2 x \, dx + \int_0^{\pi} \sin^4 x \, dx = -3 \times \frac{\pi}{2} + \frac{3\pi}{8} = \boxed{-\frac{9\pi}{8}}.$$

Otherwise, the following trick could make things a bit easier.

The integral can be divided into two parts:

$$-\int_{\pi}^{3\pi} \sin^2 x \, dx + \int_{3\pi}^{4\pi} (\sin^4 x - \sin^2 x) \, dx$$

The first term is

$$-\int_{\pi}^{3\pi} \frac{1 - \cos(2x)}{2} \, dx = -\int_{\pi}^{3\pi} \frac{1}{2} \, dx = -\pi.$$

The second term is

$$\begin{aligned} \int_{3\pi}^{4\pi} \sin^2 x (\sin^2 x - 1) \, dx &= -\int_{3\pi}^{4\pi} \sin^2 x \cos^2 x \, dx \\ &= -\int_{3\pi}^{4\pi} \left(\frac{\sin(2x)}{2} \right)^2 \, dx = -\int_{3\pi}^{4\pi} \frac{\sin^2(2x)}{4} \, dx \\ &= -\int_{3\pi}^{4\pi} \frac{1 - \cos(4x)}{8} \, dx = -\int_{3\pi}^{4\pi} \frac{1}{8} \, dx = -\frac{\pi}{8}. \end{aligned}$$

Thus, the answer is $-\pi - \frac{\pi}{8} = \boxed{-\frac{9\pi}{8}}$.

Afterword.

Thank you again for allowing me to submit a question to the HIB!

I wanted to make a question not about niche integral knowledge, but about “who can recognize patterns faster.” The question gives you some chances to perform little tricks and reduce time. I hope the two contestants go neck-and-neck and hype up the competition!

I’ve edited the question many times to find the right difficulty, and found many interesting equations in the process. For example,

$$\int_0^{\infty} \sum_{k=\lfloor \frac{x}{2} \rfloor}^{\lfloor x \rfloor} \frac{\pi}{2} \cos\left(\frac{k^2 \pi}{2} x\right) \, dx = \pi - 1 + \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} - \cdots = \pi - G,$$

where G is **Catalan’s constant**.

If you want me to change the difficulty of the problem, or submit another problem, please let me know!